

Module 6: Assignment

Security assessment and vulnerability management

MCY 630 Security Architecture

Assignment Submission Instructions

Submission requirements: assignment

File format: MSWord or pdf (preferred)

Report requirements

- Include your name and assignment title at the top of the document.
- Number the questions and answers and answer each question in your submission in sequence.
- For each response, include the required text as well as any necessary images and diagrams.
- Use black font color for texts
- If uploading an MSWord format file, view the uploaded file from Canvas to make sure that the layout of images and texts are not overlapping in the report. If the contents seem to have moved around, upload a pdf instead of an MSWord file.

Questions

1. (30 points) **Exploring CSET**
 - You will use the CSET tool to perform some security assessment tasks. You may assume any “imaginary” institution that you are assessing and respond to the questions with reasonable variations and assumptions. The objective of this exercise is to see how CSET organizes the inputs and outputs to generate the assessment report.
 - Download and install the CSET standalone application
 - URL: <https://cset-download.inl.gov/>
 - Requires: Windows OS, Download information (Org Type: *State*, Sector: *IT*, Industry: *Other*, NAISC Code: *611310*)
 - *Note:* The CSET tool is updated regularly, and the interface may change with each version.
 - Read all the instructions below and choose the options accordingly to fulfil the requirements as necessary.
 - **(Assessment 1)** Start a new assessment
 - Use the search bar to find “Cybersecurity Maturity Model Certification”
 - Name your assessment: “MCY630 CMMC” (fill other fields as necessary)
 - Fill necessary demographics fields with reasonable assumptions
 - Select “Level 5” for the CMMC assessment target
 - Complete the remainder of the assessment with reasonable variations and assumptions. Note that each aspect must be expanded with the drop-down arrow to view/answer the assessment questions.
 - In the final “Results” tab, go through the generated documentation and study the results of your assessment.
 - Submission: *CMMC Executive Summary*
 - Open the “*CMMC Executive Summary*” from the list of your reports
 - Use the “*File > Print*” and then “*Print to PDF*” option to generate the PDF for only the “*CMMC Executive Summary*”.
 - Submit only the *CMMC Executive Summary* pdf
 - Click on the “CSET” logo (top left corner) to return to the home dashboard.
 - **(Assessment 2)** Start a new assessment.
 - Use the search bar to find “Network Diagram/Components Based Assessment”
 - Name your assessment: “MCY630 NBA” (fill other fields as necessary)
 - Fill necessary demographics fields with reasonable assumptions
 - Configure SAL as follows: *Overall SAL/Confidentiality/Integrity/Availability: Moderate*
 - Select “Create a Network Diagram”
 - You will be provided with some standard network architecture options.
 - Select “Building Access Control”
 - Make any changes if you need (not necessary) and select “Return to CSET”

- Once back in the main window, you will notice that the selected network model will not populate the “Component Inventory”, which allows you specify and modify the various elements from the system architecture.
 - You are not required to make any changes in the list, but feel free to explore the items and options.
 - Move on to the assessment activity (the “Next” button in above the “Component Inventory”)
- Complete the remainder of the assessment with reasonable variations and assumptions. Note that each aspect must be expanded with the drop-down arrow to view/answer the assessment questions.
- In the final “Results” tab, go through the generated documentation and study the results of your assessment.
- Submission: *Site Summary Report*
 - Open the “*Site Summary Report*” from the list of your reports
 - Use the “File > Print” and then “Print to PDF” option to generate the PDF for only the “*Site Summary Report*”.
 - Submit only the *Site Summary Report* pdf
- Click on the “CSET” logo (top left corner) to return to the home dashboard.
- **(Assessment 3)** Start a new assessment.
 - Use the search bar to find “CISA Ransomware Readiness Assessment (RRA)”
 - Name your assessment: “MCY630 RRA” (fill other fields as necessary)
 - Fill necessary demographics fields with reasonable assumptions
 - Complete the remainder of the assessment with reasonable variations and assumptions. Note that each aspect must be expanded with the drop-down arrow to view/answer the assessment questions.
 - In the final “Results” tab, go through the generated documentation and study the results of your assessment.
 - Submission: *RRA Report*
 - Open the “*RRA Report*” from the list of your reports
 - Use the “File > Print” and then “Print to PDF” option to generate the PDF for only the “*RRA Report*”.
 - Submit only the *RRA Report* pdf
- **(Submissions)** Upload only the following files for the CSET assessments completed above with the following filenames:
 - *CMMC Executive Summary.pdf*
 - *Site Summary Report.pdf*
 - *RRA Report.pdf*

2. (10 points) Exploring Microsoft TMT

- Download and install the Microsoft TMT standalone application
 - URL: <https://www.microsoft.com/en-us/securityengineering/sdl/threatmodeling>
- Follow the “Getting Started” online guide to build a sample model (optional: feel free to add items to your DFD)
- Follow the “Feature Overview” online guide and explore Element Properties, Messages, and Analysis View options.
- Finally, use the report generation option to export the threat modeling report
- **(Submission)** Upload only the exported threat modeling report completed above with the following filename:
 - *Microsoft TMT Report.pdf*

3. (5 + 5 points) Task Report

Create a task report to summarize your actions for the above two tasks. Include a reflection on the application usage, how you feel these tools could be beneficial, the type of information you could extract from the reports, and your overall experience with the activity.

Submit the task report (*Task Report.pdf*) along with the above 4 documents.